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Determination of the atomic nitrogen flux from a radio frequency plasma nitride source for molecular beam epitaxy systems

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Direct measurements of both the dissociation fraction and the atom flux have been carried out for the first time to characterize a radio frequency (rf) atomic nitrogen source. The measurements have been made over a range of source pressures and rf powers. Dissociation fractions of up to 0.4 and atomic nitrogen fluxes of up to 0.85×10^{18} atom s^{-1} were obtained. In addition these results have been correlated with spectroscopic measurements of the ratios of atomic and molecular optical emissions from the source plasma in order to provide a way of estimating *in situ* the atomic flux.

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I. INTRODUCTION

There is growing interest in the use of atomic nitrogen sources for material deposition and for the modification of surface properties. The most widely employed source in molecular beam deposition (MBE) systems for nitrogen doping and the growth of semiconductors is the rf plasma source.¹

A qualitative indication of the performance of such atomic sources has been obtained using spectroscopic techniques. The atomic and molecular emissions are monitored to provide an indication of the efficiency of the source and of the atomic nitrogen flux.² In these measurements the fraction of dissociated molecules D and the atom flux F in the emergent beam is not measured directly. In the present work a direct measurement of both the dissociation fraction and the atomic nitrogen flux from an SVT Associates (SVTA) rf plasma source (model RF 4.5) was made using a technique previously developed in the laboratory of The Queen's University in Belfast.³ In addition these results were correlated with spectroscopic measurements of the ratio of atomic and molecular emission intensities in a manner similar to our previous work.⁴

II. EXPERIMENTAL APPROACH

A. Description of the source

The source is described in detail elsewhere⁵ so only the essential features of it are given here.

The SVTA RF 4.5 is an ultrahigh vacuum (UHV) compatible source mounted on a conflat NW63 flange. The source consists of a pyrolytic boron nitride (PBN) cylindrical plasma tube, 133.6 mm in length and 16.8 mm in diameter, surrounded by a water-cooled rf coil operating at 13.56 MHz. The rf power operating levels are in the 200–500 W range. Nitrogen is introduced at one end of the plasma tube and the partially dissociated nitrogen beam effuses from a

2.8 mm aperture at the other end. A manually tuned rf matching network is employed to minimize the reflected power.

B. Measurement of the dissociation fraction

A schematic diagram of the experimental system is shown in Fig. 1.

The source was mounted on a high vacuum chamber and fed via an UHV gas control valve with pure nitrogen (research grade 99.998% purity) at constant pressure P_S measured close to the source inlet by a temperature controlled capacitance manometer (MKS baratron type 170M-6B). A small portion of the partially dissociated beam effusing from the 2.8 mm diam aperture into the high vacuum chamber entered a second differentially pumped chamber. This beam

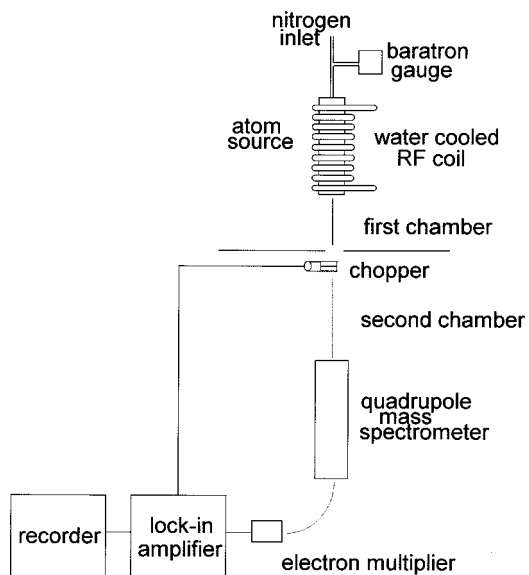


Fig. 1. Schematic diagram of the experimental setup.

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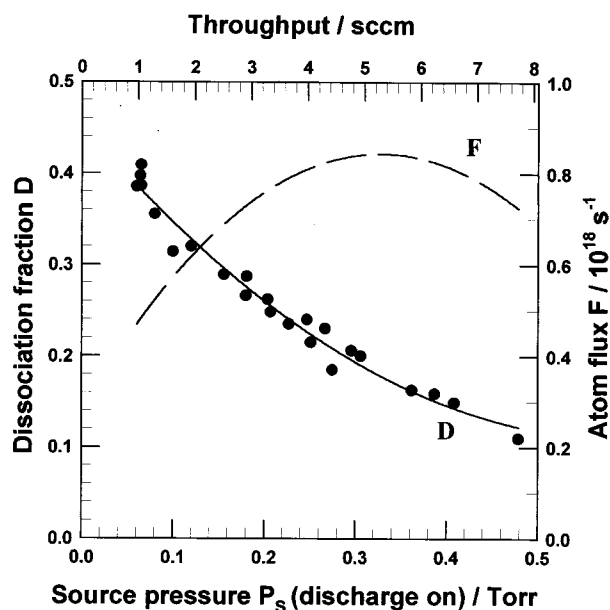


FIG. 2. Dissociation fraction D and atom flux F as a function of the indicated source pressure/throughput for a rf input power of 500 W.

was modulated by a electromechanical tuning-fork type chopper at a frequency of 180 Hz and its molecular content was analyzed by a quadrupole mass spectrometer which was tuned to transmit molecular nitrogen ions. The N_2^+ ion beam leaving the mass spectrometer was then detected by an electron multiplier (Thorn EMI type 9642/2) mounted 90° with respect to the quadrupole axis and the signal was amplified by a phase sensitive detection system (EG&G model 5101). The modulation and phase sensitive detection were necessary to distinguish ions in the beam from the background gas. The molecular nitrogen signal was recorded, and the dissociation

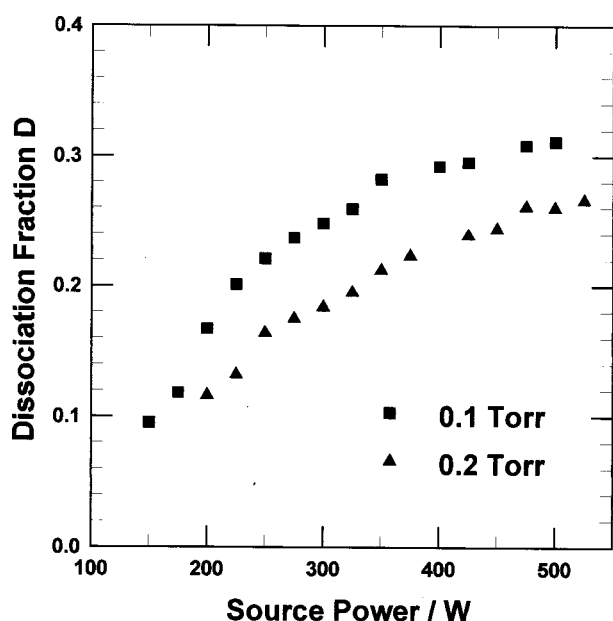


FIG. 3. Dissociation fraction D as a function of rf power at two indicated source pressures P_s .

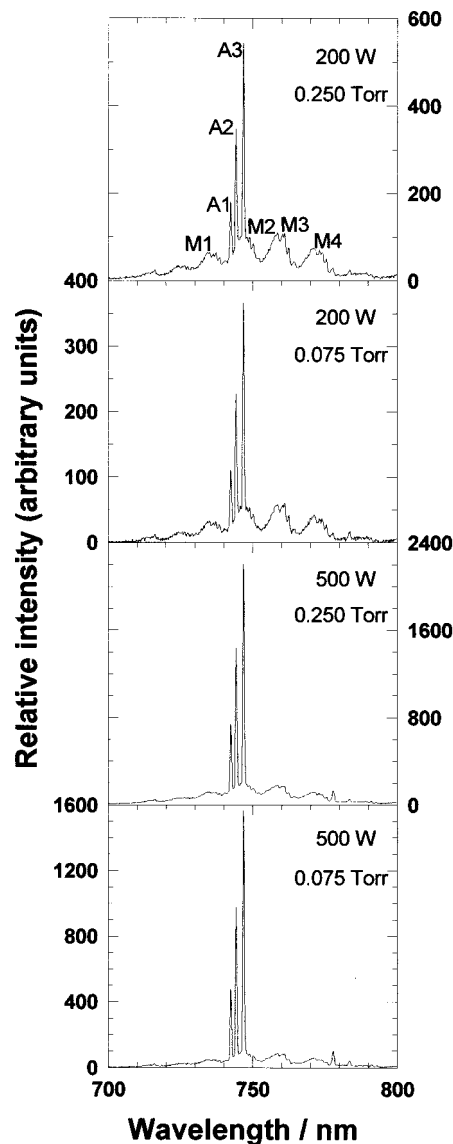


FIG. 4. Optical emission spectra from the nitrogen plasma at two different indicated source pressures and rf power levels of 200 and 500 W.

fraction D was determined by comparing the N_2^+ ion signal intensity with the plasma ignited (S_1) and extinguished (S_2) using the relation

$$D = (S_2 - S_1) / S_2. \quad (1)$$

Signal S_2 was measured immediately after the plasma extinguished when the source temperature was close to that when the plasma was ignited.

C. Gas throughput and atom flux

The gas throughput Q was measured for specified source pressures P_s ranging from 0.1 to 0.6 Torr by monitoring the rate of change of the pressure P in a cylinder of volume V feeding the source. The time Δt taken for the pressure P to fall by ΔP was measured at a constant source pressure P_s . The throughput was derived from the relation

$$Q = V \Delta P / \Delta t \quad \text{in Torr l s}^{-1}. \quad (2)$$

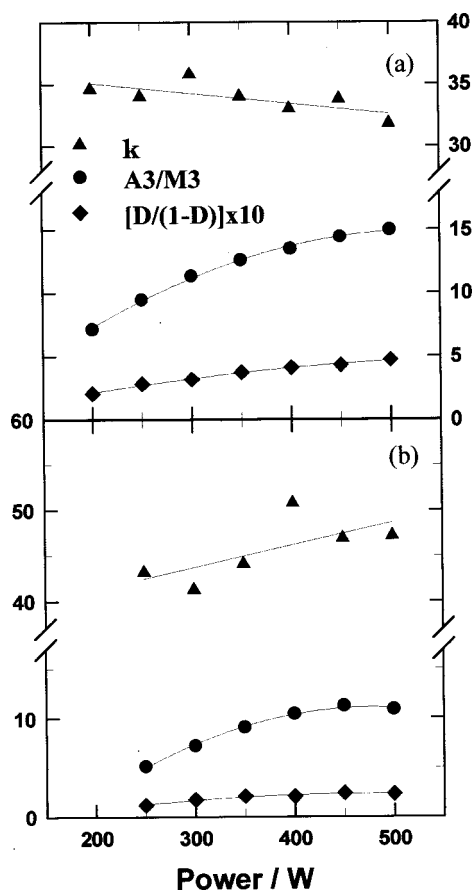


FIG. 5. A3/M3 and $D/(1-D)$ (see the text for further discussion) as a function of power at (a) 0.1 and (b) 0.3 Torr.

Absolute measurements of the gas throughput and the associated dissociation fractions enabled the nitrogen atom flux F leaving the source to be determined using the relation

$$F = 3.22 \times 10^{19} 2DQ, \quad (3)$$

where F is in atom s^{-1} and Q is in Torr l s^{-1} at 300 K.

D. Optical measurements

Optical emissions from the plasma were recorded and computer analyzed by an Oriel Instruments MultispecTM spectrometer fitted with a 1200 line/mm grating and linked to the source via a fiberoptic connection. A complete description of the spectral measurement can be found in Ref. 4. The spectrometer was set for a spectral region, 700–800 nm, where both atomic and molecular emissions can be observed simultaneously.

Assuming that (a) the atomic line and molecular band emissions are due to direct plasma electron excitations of the N atoms and N_2 molecules and (b) that the cross sections for the selected excitations follow a similar energy dependence, the following relation holds:

$$\frac{A}{M} = k \frac{D}{1-D}, \quad (4)$$

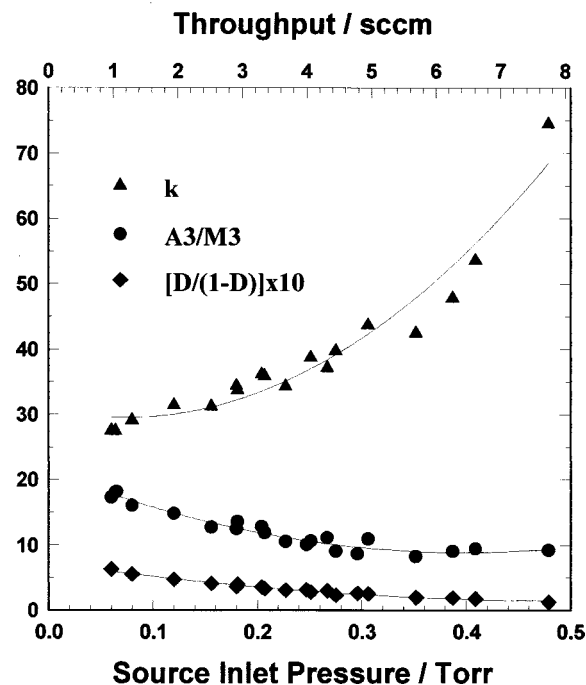


FIG. 6. A3/M3 and $D/(1-D)$ (see the text for further discussion) as a function of the indicated source pressure/throughput for a rf input power of 500 W.

where A and M are selected atomic line and molecular band emissions, respectively, k is a constant, and $D/(1-D)$ is proportional to the ratio of atomic to molecular density in the source plasma.

III. RESULTS AND DISCUSSION

Figure 2 shows the dependence of the dissociation fraction D and the atomic flux F as a function of indicated source pressure P_s at an applied rf power of 500 W. Dissociation fractions were found to decrease from 40% at 0.07 Torr to about 10% at 0.50 Torr. The corresponding atom flux ranged from 0.45×10^{18} up to 0.85×10^{18} atoms s^{-1} with a maximum at 0.32 Torr.

Figure 3 shows the dependence of the dissociation fraction as a function of source power for indicated source pressures of 0.1 and 0.2 Torr.

For both source pressures D was found to increase with the applied rf power reaching a maximum value of 30% for 0.1 Torr and 26% for 0.2 Torr at 500 W.

Figure 4 shows several high resolution spectra of the source plasma taken under various source operating conditions. The atomic triplet of wavelengths 742.4 (A1), 744.2 (A2), and 746.8 nm (A3) is associated with the $3s^4P-3p^4S^0$ transition,⁶ and the molecular bands, with band heads at 738.7 (M1), 750.4 (M2), 775.3 (M3), and 872.2 nm (M4), are associated with the $B^3\Pi_g-A^3\Sigma_u$ transitions.⁷ It can be seen that while the overall intensity of the spectrum increases with increasing pressure, the relative intensity of the atomic spectrum compared with the molecular spectrum decreases. However, when the power is increased, both the overall intensity and the ratio of atomic to

molecular emission intensities increase. The ratio of atomic to molecular emission intensities A_3/M_3 has been compared with $D/(1-D)$ as a function of power (Fig. 5) and pressure (Fig. 6).

As can be seen in Fig. 5, a good correlation was found between the ratio of atomic to molecular emission intensities A_3/M_3 and the ratio $D/(1-D)$ as a function of power. There is reasonable agreement with Eq. (4) particularly at 0.1 Torr, with a value for k close to 34. The agreement is also reasonably good at 0.3 Torr but with a larger k value of approximately 45. The results shown in Fig. 6 demonstrate that the factor k does not remain constant but is a function of pressure increasing with increasing source pressure.

IV. CONCLUSION

The SVTA rf plasma source was found to produce nitrogen beams with a degree of dissociation of up to 40% at 0.1 Torr and an atomic flux of up to 0.85×10^{18} atom s^{-1} at 0.32 Torr. Spectroscopic measurements of the plasma emissions have been correlated with measurements of the dissociation fractions. These data can be used, in combination with the measured gas throughput, to determine the atomic nitrogen flux leaving the source exit aperture.

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